



NeuroEye

NEURO-OCULAR INTERFACE DRIVEN INTELLIGENT ELECTRO-MECHANICAL DEVICE

Using EEG & EOG Signals for Assistive Mobility

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PROBLEM STATEMENT

- Individuals with mobility impairments struggle to use traditional assistive devices that require physical control (joysticks, switches, buttons).
- Limited hand or body movement reduces independence and makes daily mobility difficult.
- There is a need for a hands-free, intuitive control system that works without physical effort.
- Combining BCI (brain signals) and EOG (eye movements) can overcome these limitations by enabling users to control devices mentally and visually.



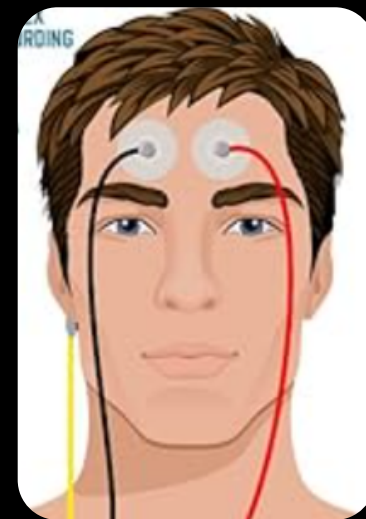
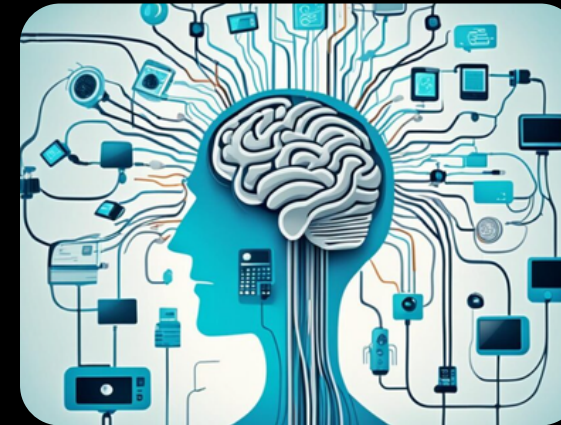
INTRODUCTION

NeuroEye is a system that lets users control a toy car using brain signals (EEG) and eye movements (EOG).

Brain–Computer Interface (BCI)

A technology that creates a direct link between the brain and an external device.

It allows users to control machines using brain activity—without physical movement.

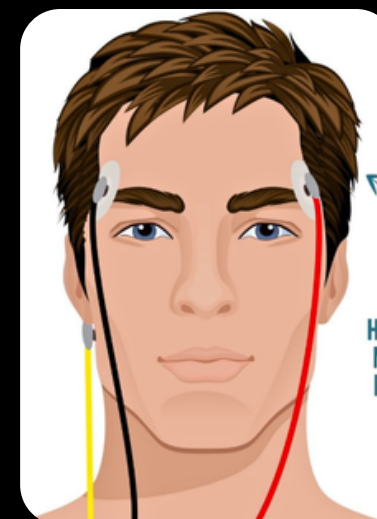


Electroencephalography (EEG)

A method used to record electrical signals produced by the brain.

In NeuroEye:

→ EEG is used to detect commands like Forward and Stop.



- **Electrooculography (EOG)**

A technique that measures the electrical activity produced by eye movements.

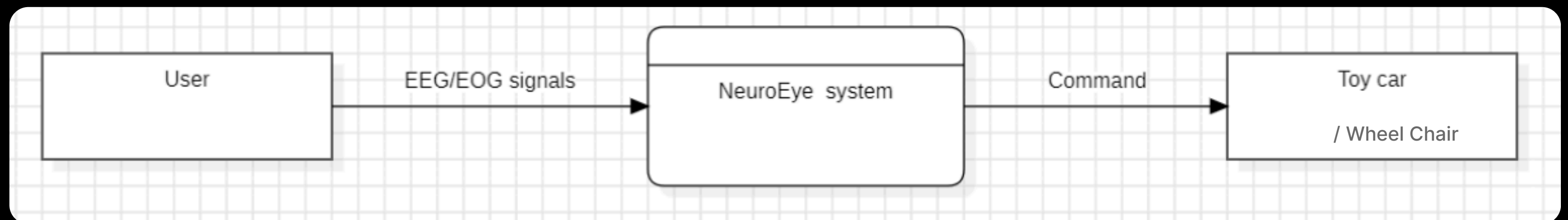
In NeuroEye:

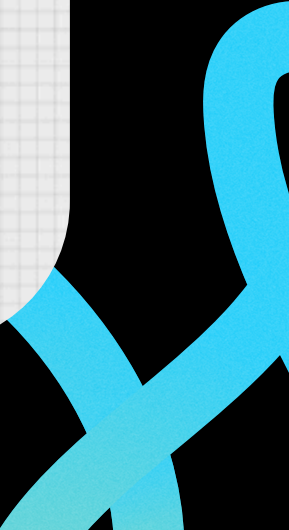
→ EOG is used to detect Left and Right eye movements.



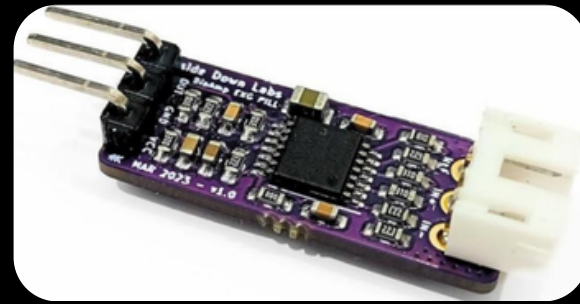
PROPOSED SOLUTION

- NeuroEye provides a hands-free control system using brain and eye signals.
- EEG detects brain activity to control Forward and Stop.
- EOG detects eye movements to control Left and Right.
- Signals are captured using the BioAmp EXG Pill and processed by an Arduino.
- Removes the need for physical controls like joysticks or switches.
- Enables people with limited mobility to control devices easily and independently.





HARDWARE REQUIREMENTS



Arduino Uno Microcontroller

BioAmp EXG Pill (2)

BioAmp cables

Electrodes

Jumper cables

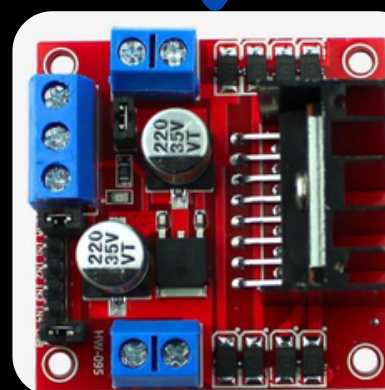
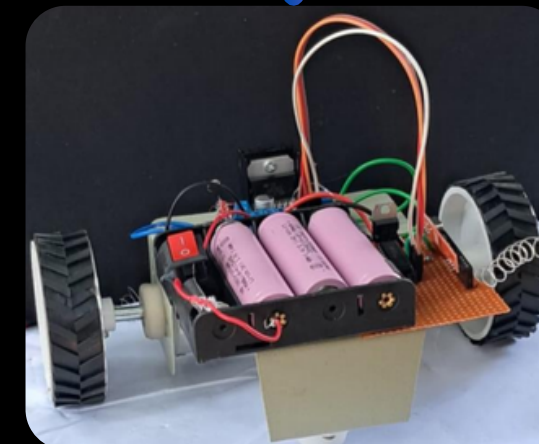
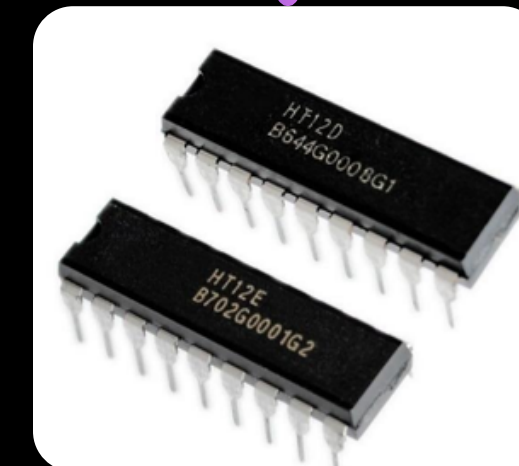
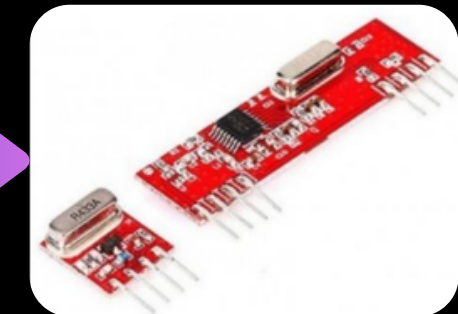
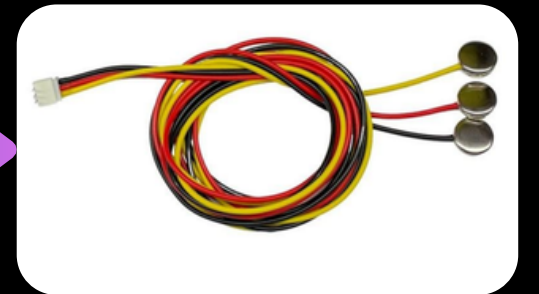
Conductive gel

RF Module

Motor Driver

HF12D (Decoder), HT12E(Encoder)

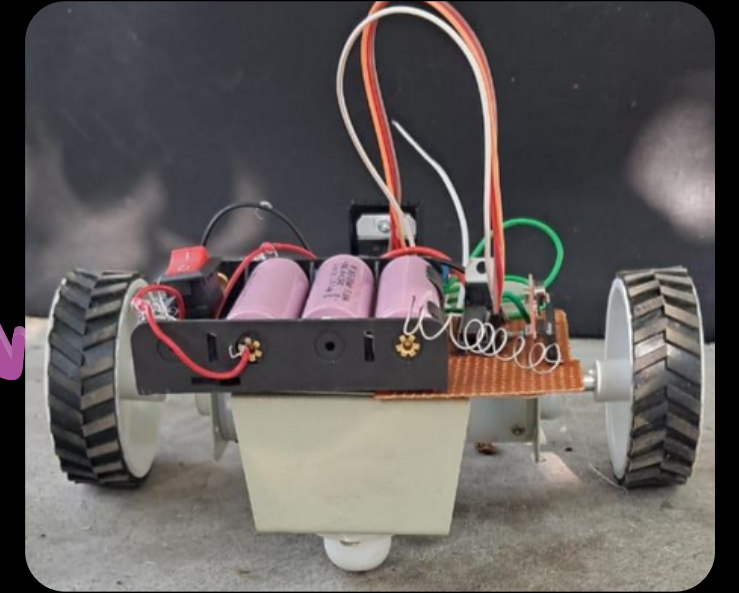
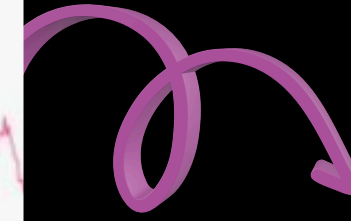
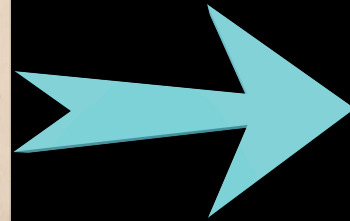
Toy car



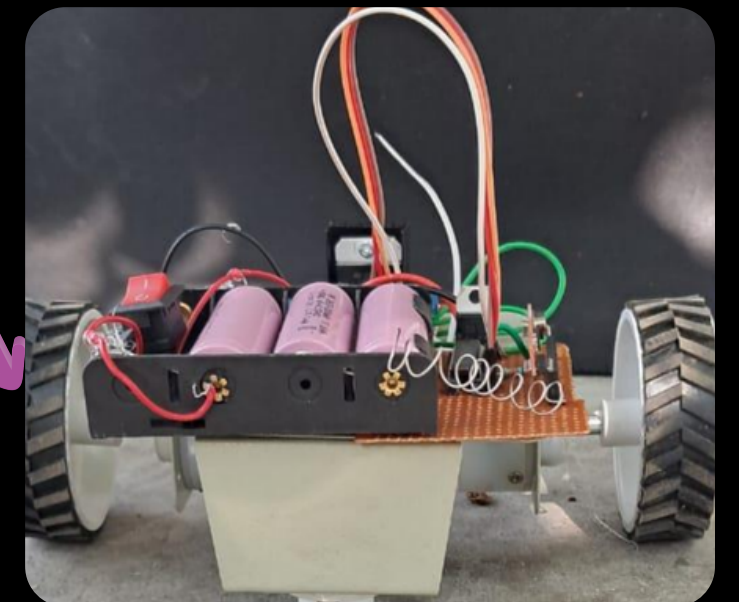
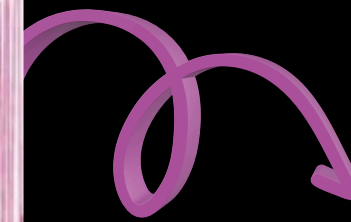
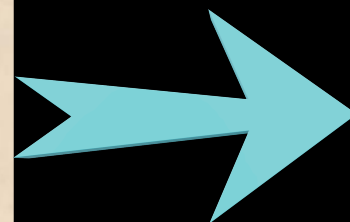
OUTCOME



EEG SIGNAL



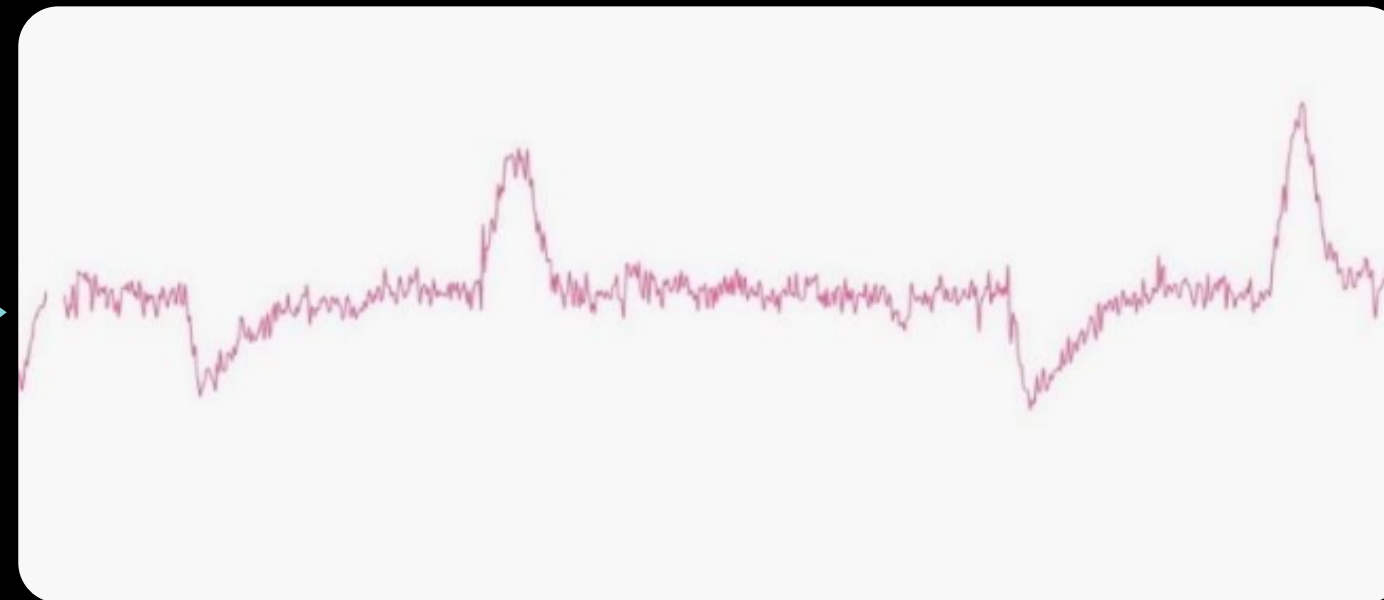
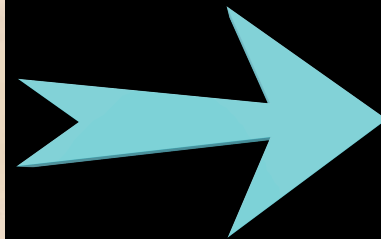
"Stop" Command (Relaxed State)



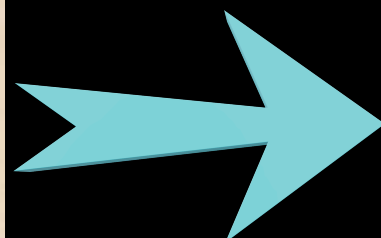
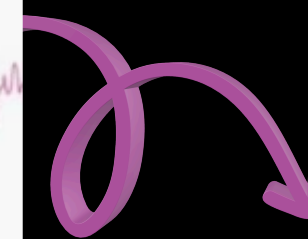
"Forward" Command (Concentrate state)

OUTCOME

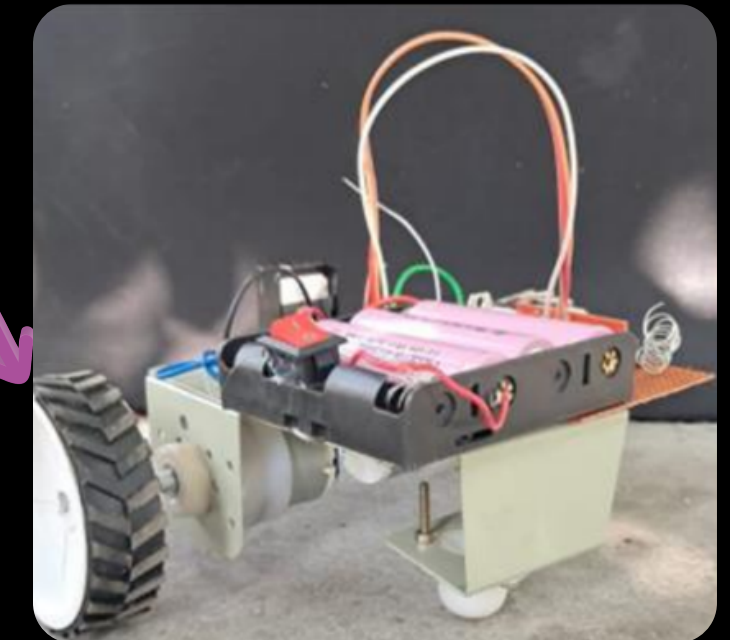
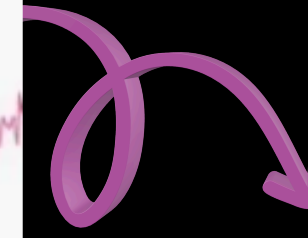
EOG SIGNAL



"Left" Command (Left Horizontal Eye Movement)



"Right" Command (Right Horizontal Eye Movement)



FUTURE SCOPE

- **Obstacle and Object Detection**

Integration of sensors such as ultrasonic sensors, LiDAR, or cameras to detect obstacles and ensure safe navigation by automatically avoiding collisions.

- **Route Mapping and Destination Navigation**

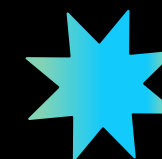
Implementation of indoor route mapping that allows users to select predefined locations such as bathroom, bedroom, or living room. The system will automatically calculate and follow the optimal path.

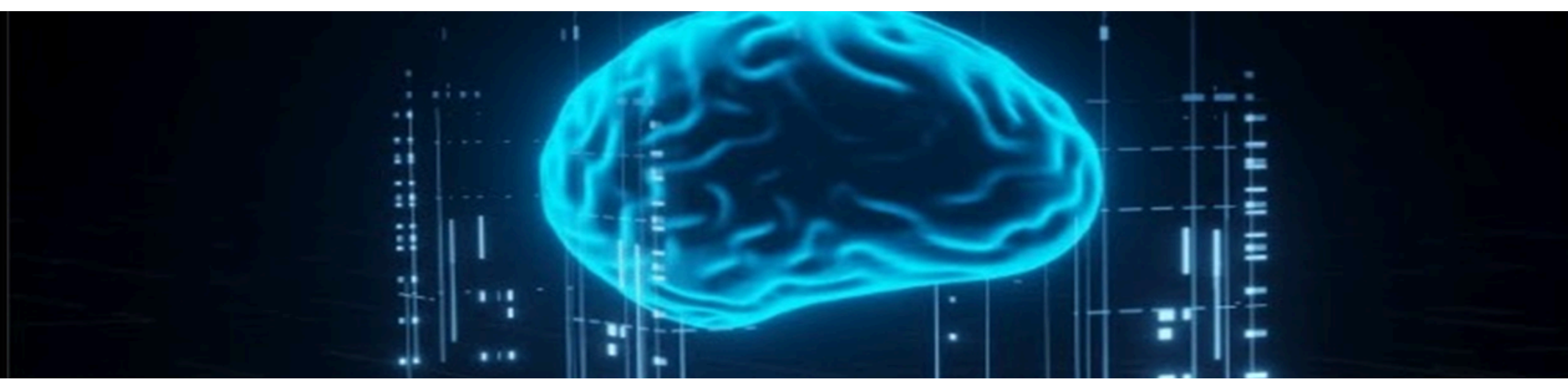
- **SLAM-Based Autonomous Navigation**

Use of Simultaneous Localization and Mapping (SLAM) to create real-time maps of the environment and determine the device's position, enabling intelligent autonomous navigation.

- **Fall Detection and Emergency Alert System**

Integration of accelerometers or posture sensors to detect falls or abnormal conditions and automatically send alerts to caregivers or emergency contacts.





**CONNECTING
OUR BRAIN TO
THE INTERNET**

THANK YOU

With NeuroEye, we imagine a world where mobility is driven by
thought, not limitation.
Let's innovate toward a future where technology uplifts every
individual.

